



PROGRAMME
DE RECHERCHE
MATÉRIAUX
ÉMERGENTS



AI data bases and Material Discovery

E. Farhi, *EXP/GRADES*

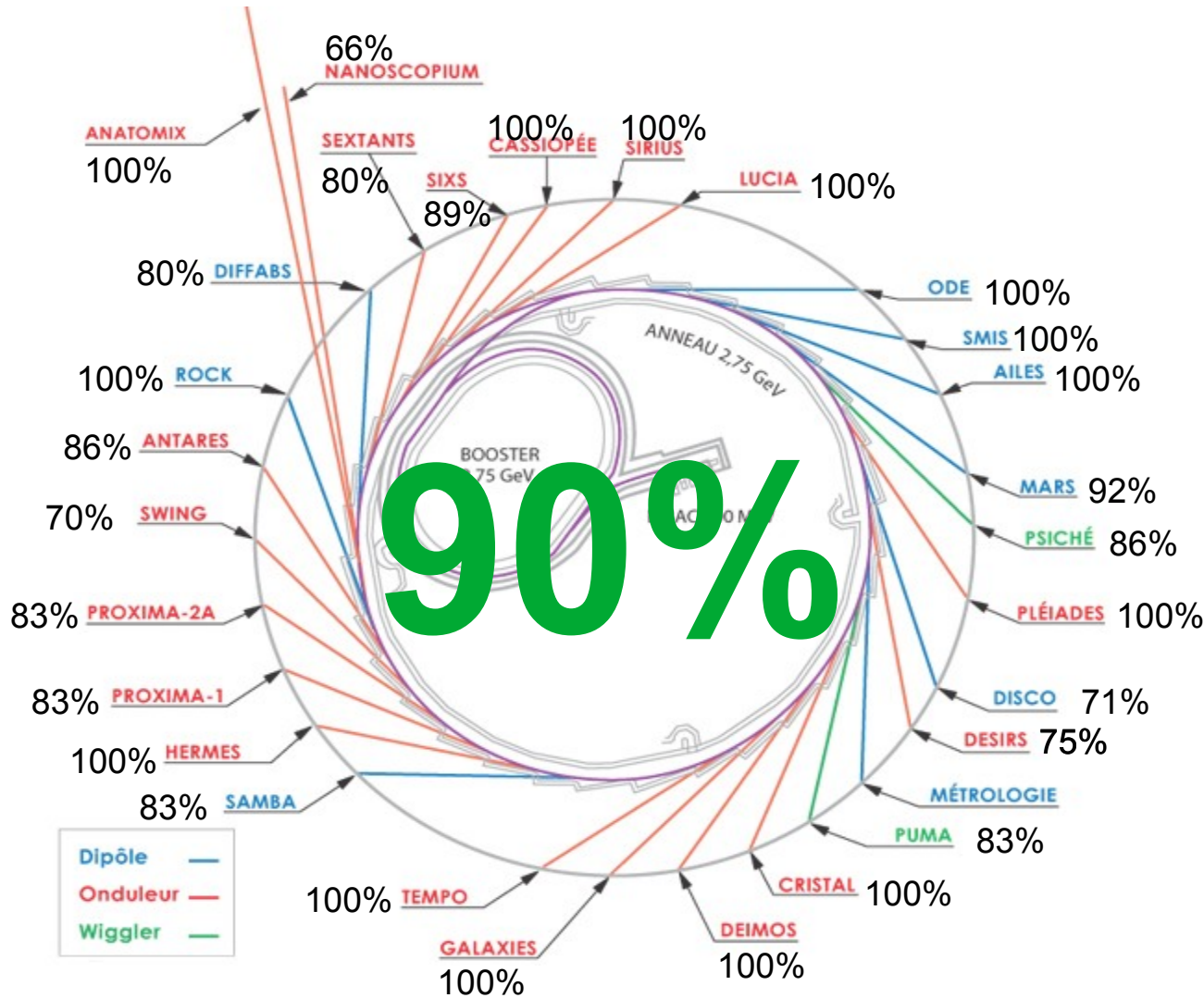
DIADEM School Grenoble 2026

Data reduction and Analysis Group at SOLEIL

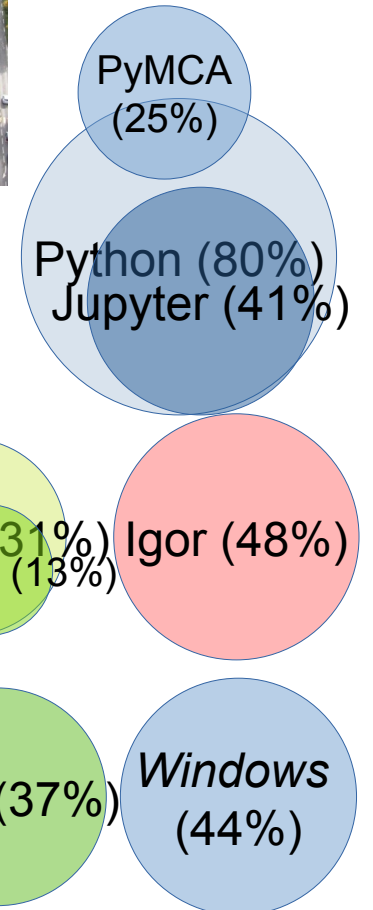
- **Groupe de Réduction et d'Analyse de Données Expérimentales de Soleil (GRADES).**
- Created in Feb 2020.
- **Staff:** 6 perm. (3 dev, 2 BL, 1 AI) + ~4 temp.
 - E. Farhi, F. Picca; O. Roudenko, S. Bac; A. Bellachehab, G. Cassam
- **Topics:** scientific software (diff, spectro/abs, simulation, ...), Debian/Ubuntu packaging, deployment, coding, documentation, support, ... and now AI everywhere.
- **Means:** 3 central servers + ~25 BL servers.



Beam-line requirement capture



29 BLs
 ~100 "crazy" scientists
 160 requirements
 144 supplied (90%)
 4000 visitors/y
 700 publications/y
 data: 3 PB/y
 Budget 61 M€/y



Beam-line requirements: software

Our BLs require essentially :

- **Quasar/Opus PyHST2 UFO ImageJ Fiji Napari Nabu Astra-Toolbox *matlab* silx navarp arpys binoculars pynx labplot jupyterhub jupyterlab codeblock vscode pyfai pymca maud athena/artemis corali-q4 cryspy/quanty fdmnes quantum-espresso/sssp abinit cp2k gromacs lammps fastosh xraylarch sasview cctbx dioptas bornagain genx anarod xrayutilities ase foxtrot atsas bioxtas-raw matlab-csaxes-ptycho**
- *Igor RefFIT casaxps* (Windows)

We develop specific software:

- **binoculars/hkl, McXtrace, pyfai/pynx, qemu-web-desktop, notebooks, BL data I/O, ...**



debian

- We have chosen **Debian-class** systems to ensure **coherence**, **stability** and **maintainability** of most of our scientific software.



debian

- The Debian Science stack is **extensive**.
- Brings better life-span than **conda** and **containers**.
- We use 'unstable' packages on request.



- Two Debian contributors in the group + 40k€/y subcontracting.



~**400** packages pushed in Debian Science over 4 years.

<https://salsa.debian.org/pan-team/soleil-packaging-overview>

F. Picca
E. Farhi
R. Mas
X. Guimard

Load raw data (handle formats).

Normalize/clean/correct intensity:

$$\frac{\text{Light} - \text{Dark} - \text{Offset}}{\text{Flat} - \text{Dark} - \text{Offset}}$$

Align (image registration) and **Stack**.

Transform coordinates. Add metadata.

We mostly use Python libraries:

- `pyfai`, `pymca`, `silx`, `h5py`, `libxrl`,
`xraydb`, `larch`, `pyhst/nabu`, `skimage`,
`sklearn`, `orange3/quasar`, `pynx`, ...



Load reduced data.

Make use of **metadata**.

Use **large data sets**, #CPU and #GPU.

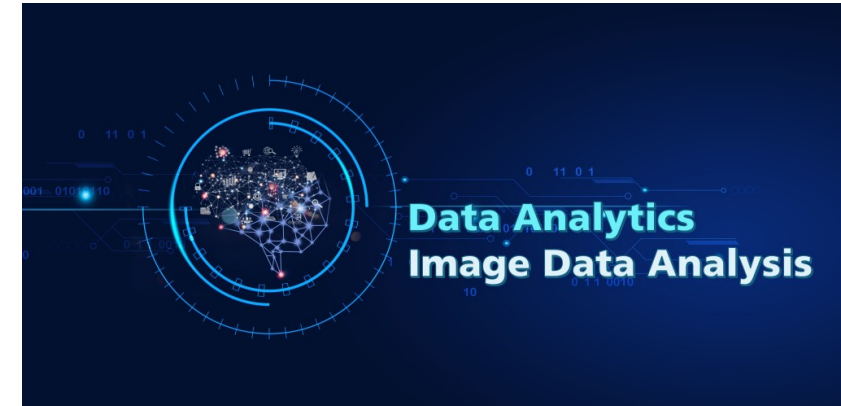
Usually perform **statistics**.

Fit theory (extract physical parameters).

Estimate **errors**.

Write scientific **papers**.

- Dioplas, Mantis-Xray, Larch, Arpys/Navarp, Orange3/Quasar, BioXtas-RAW, SASView, BornAgain, GenX, Napari, Fiji, ...



We provide all inclusive Data Processing services:



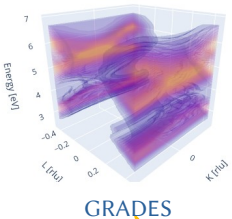
DARTS (`apt install qemu-web-desktop`).

Thousands of software and libraries. <https://data-analysis.synchrotron-soleil.fr/desktop>



JupyterHub (shared storage with *DARTS*).

<https://data-analysis.synchrotron-soleil.fr/notebook/>



Material modelling (easy DFT+ e^-/ph dispersions)

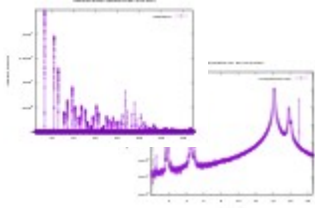
<https://data-analysis.synchrotron-soleil.fr/material-modelling/>



Chat AI (Gemma4 – use on-site Knowledge Base)

Diffraction + Fluorescence estimates

https://data-analysis.synchrotron-soleil.fr/computing/fluor_diffraction.html



Data analysis: DARTS



Service: <https://data-analysis.synchrotron-soleil.fr/>

Code: <https://gitlab.com/soleil-data-treatment/soleil-software-projects/gemu-web-desktop>

Data Analysis Remote Treatment Service (DARTS)



This service is a data analysis portal that allows to create a remote desktop to treat your data, in the cloud. You can tune the type of system you need. It will be displayed in your browser, without any additional software for you to install.

Read more on our [HELP](#) page.

NOTE: From SOLEIL Network, please use Firefox with "auto-detect" proxy (Top-right menu, Preferences, search for "proxy" (top right), select Network Settings, choose "auto-detect"). It is also important to inactivate any JavaScript blocker plugin.

~350 users/month

1

User ID: farhie

Password:

Create

By pressing the Create button, you agree with our Terms and Conditions (*)

Machine: Data Analysis (Debian 11, stable, "Ute...")

Configuration script (opt.): Path or URL to a single file or 'exec: cmd1; cmd2; ...'

Number of CPU's: 4

Amount of memory: 4 GB

Compute on GPU (opt.): ☐

You may request a physical GPU to e.g. run heavy computations (not for display). The tools and libraries you wish to use should have been designed to benefit from such devices with e.g. OpenCL, CUDA, OpenACC, ROCm/HIP. Do NOT request a GPU if you do not actually use it, as their number is limited.

Fully configurable.
Low Tech / Maintenance.
Deployment in **minutes**.
Scalable.
DARTS is simple and efficient.

Currently Includes:

- CP2K, GROMACS, LAMMPS, QE/SSSP, ABINIT, ASE, Jupyter NB, FEFF/FDMNES, ORCA, FPLO, PyTorch, Sk-Learn/image...

Data Analysis Remote Treatment Service: debian-bookworm-grades-2026-05-29.qcow2

OK You can use your session until Wed Sep 2 21:00:11 2026.

Hello farhie !

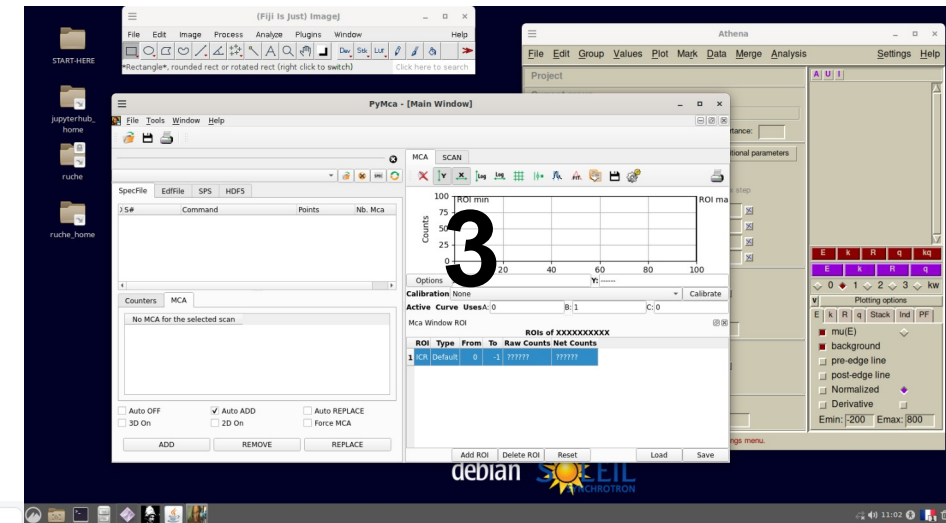


2

Connect Share Stop

NOTES:

- You can close the browser and reconnect any time (within life-time) with the link above.
- Select the [Manage sessions] item in the service login page to list, reconnect or abort your sessions.



QR code.
Access to **GPUs**.
Open-source
100+ of sci software in VM.
Access to a persistent area.
Access to the Ruche (exp data).
Will evolve into the **VISA** service.

Material dispersions modelling service

Material modelling (easy DFT+ e^-/ph disp.)

- **Service:** https://data-analysis.synchrotron-soleil.fr/computing/material_modeling.html
- **Code:** <https://gitlab.com/soleil-data-treatment/soleil-software-projects/material-modelling-service>

Input
CIF

Material Modeling Service

This service allows to compute some electronic and phononic properties of materials from first principle theory (DFT *ab-initio*). The only requirements are your credentials, and a CIF/PDB/POSCAR etc (see the [ASE documentation](#)). In addition, you may choose the calculator to use (GPAW, QE), the computational setups to perform.

You are invited to search for your material in the following resources:

- [Crystallography Open Database \(COD\)](#)
- [MaterialsProject](#)
- [NoMaD repository](#)
- [PhononDB](#)

or simply use AI for a test.

The results will be stored into our data storage at <https://data-analysis.synchrotron-soleil.fr/data/modeling>. By submitting your data here, you implicitly agree to share it with other colleagues at SOLEIL.

Your User ID *:

Your password *:

Calculator:

Smearing:

Compute:

Material structure *: Aucun fichier sélectionné.

Ground state on Wed May 28 16:16:55 2025

Generated:

- [\[properties.h5\]](#)
- Energies are e.g. in [eV].
- Refer to the [ASE](#) documentation to learn about signification and units.

Properties:

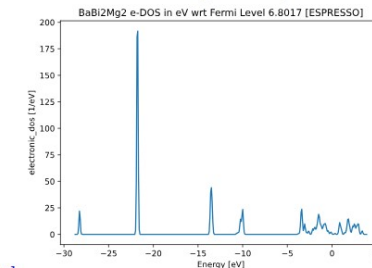
```
{ 'angular_momentum': array([0., 0., 0.]), 'center_of_mass': array([0.94611635, 1.6387216, 3.21874433]), 'charges': None, 'dipole_moment': None, 'Fermi_level': 6.8017, 'kinetic_energy': 0.0, 'magnetic_moment': None, 'momenta': array([[0., 0., 0.], [0., 0., 0.], [0., 0., 0.], [0., 0., 0.], [0., 0., 0.]]), 'moments_of_inertia': array([ 876.33055125, 4242.82588009, 4984.41961667]), 'potential_energy': -11893.656298752589, 'stress': array([-0.00443654, -0.00443654, 0.00146079, -0., -0., 0.]), 'total_energy': -11893.656298752589 }
```

Electronic density of states on Wed May 28 16:16:55 2025

Generated:

Full report and data files

- BaBi2Mg2 e-DOS in eV wrt Fermi Level 6.8017 [ESPRESSO]



- [\[electronic_dos.svg\]](#)
- Tabulated density of states data is available in [\[properties.h5/electronic_dos\]](#).

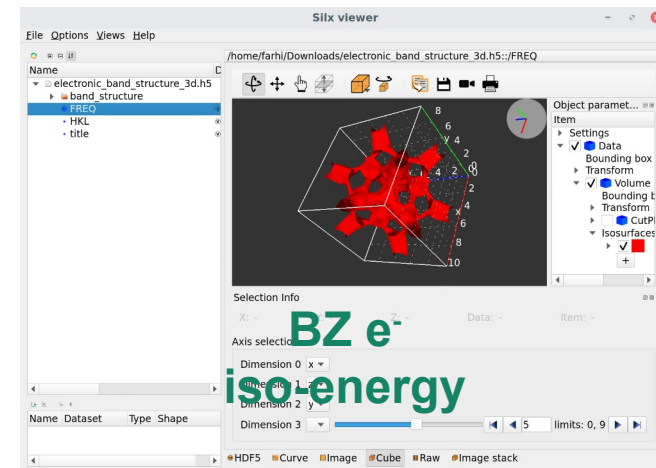
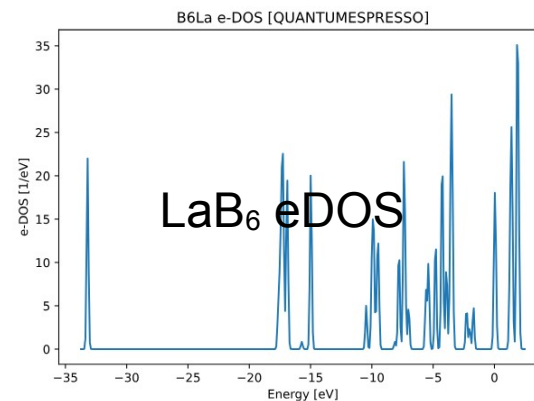
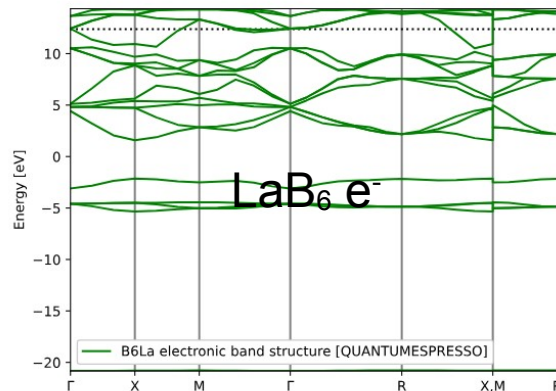
https://data-analysis.synchrotron-soleil.fr/data/material_modeling/BaMgBi2.cif-QuantumEspresso-bertran-20250528-161646/data/properties.h5

Material modelling (easy DFT+ e^-/ph disp.)

- We compute the electronic levels in the BZ using e.g. *QE/SSSP* via *ASE*.
- Configuration is automatic for fast estimates (based on *Materials Project* settings):

Calc	Conv. dE	Displ. δx	E_{cut}	Kpts	supercell
QE/SSSP	1e-5 eV	0.01 Angs	$200 \cdot N_{\text{at}}$ eV	$N_{\text{kpts}} / \sqrt[3]{(N_{\text{at}} \cdot \text{supercell})}$	$\sqrt[6]{(K_{\text{grid}}^3 / N_{\text{at}})}$

- From this we get electronic dispersions:



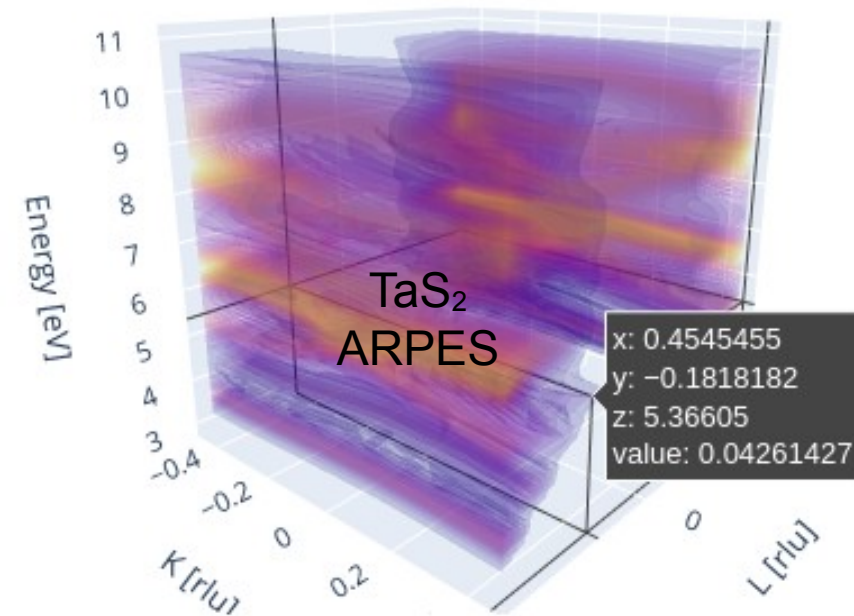
Electronic states → ARPES intensity

- From the Fermi golden rule via the electro-magnetic field $\mathbf{A}$ and light momentum $\mathbf{p}$

$$\Gamma_{i \rightarrow f} \propto | \langle f | \mathbf{A} \cdot \mathbf{p} | i \rangle |^2 \delta(E_f - E - h\nu)$$

- Field $\mathbf{A}$ boils down to photon polarization, and $\mathbf{p}$ to the in-plane momentum (surface).
- We compute an estimate for the intensity on a flat sample surface

$$I(q, E) \propto \sum_{bands} |\vec{\epsilon} \cdot \vec{q}_{||}|^2 \frac{1}{1 + e^{(E - E_F)/kT}} \frac{1}{1 + (E - E_f)^2/\Gamma^2}$$



Small displacements → vibrational states

- From the Fermi golden rule via the interatomic potential V and particle density r .

$$w_{\vec{k}_i \rightarrow \vec{k}_f, \lambda_i \rightarrow \lambda_f} = \frac{2\pi}{\hbar} |\langle \lambda, \vec{k}_f | V | \lambda_i, \vec{k}_i \rangle|^2 \rho(E_f),$$

Use *PhonoPy*

- We compute the form factors (harmonic approx – small displacements → forces and potential V)

$$F_j(Q) = \sum_a \frac{b_a}{\sqrt{m_a}} e^{-W_j(Q)} \vec{Q} \cdot \vec{e}_j e^{-i\vec{Q} \cdot \vec{r}}$$

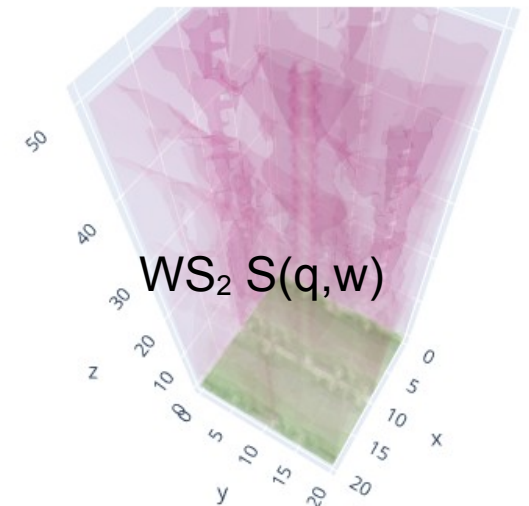
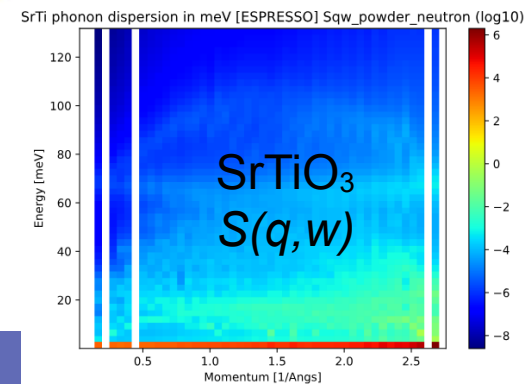
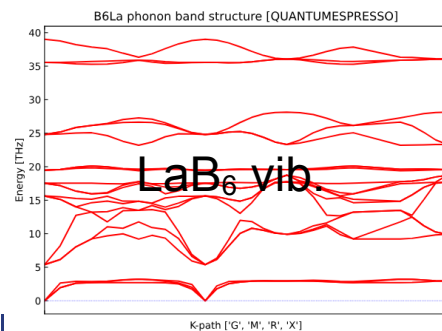
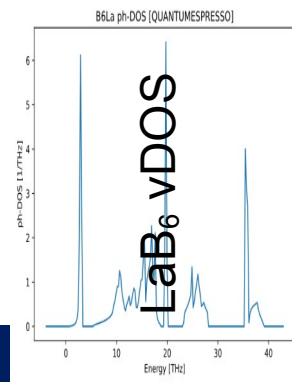
b = scatt. length [m] = $r_e f$ for photons

$$W_j(Q) = \frac{|\vec{Q} \cdot \vec{e}_j|^2}{2\omega_j(Q)m_a} (2n(\omega_j(Q)) + 1)$$

$$n(\omega) = 1/(e^{\hbar\omega/k_B T} - 1)$$

- And then the X-ray and neutron vibrational dynamic structure factors.

$$S(\vec{Q}, \omega) = \sum_j (n(\omega) + 1) \frac{|F_j(Q)|^2}{\omega_j} DHO(\omega, \omega_j, \Gamma)$$



Material fluorescence modelling service

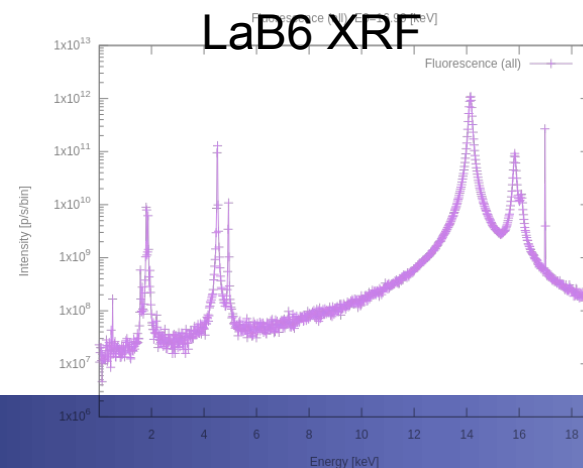
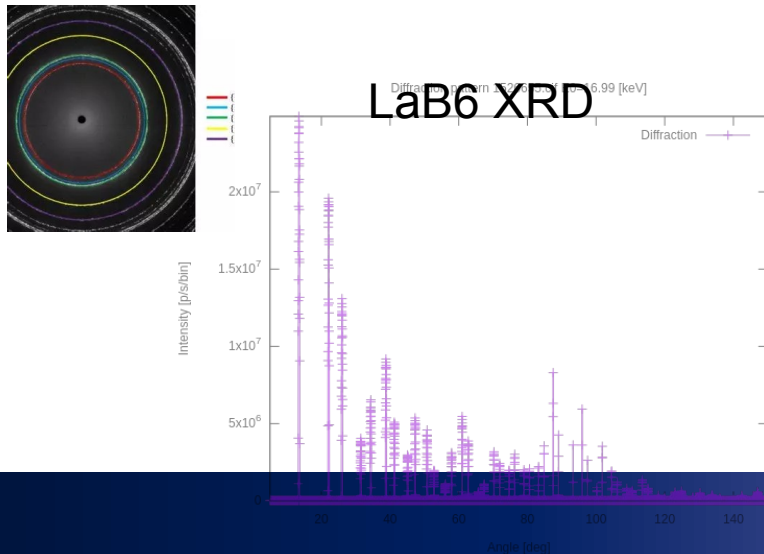
Diffraction + Fluorescence estimates

- https://data-analysis.synchrotron-soleil.fr/computing/fluo_diffraction.html
- <https://gitlab.com/soleil-data-treatment/soleil-software-projects/fluo-diffraction-service>

We rely on **XrayLib** (librxi) and **McXtrace** to compute the powder diffractogram and the fluorescence spectra. Also compute the experimental “background”. Produce *realistic* data sets.

$$\sigma_{\text{cone},Q} = j_Q f \exp(-2W) \frac{V}{V_0^2} \frac{\lambda^3}{4 \sin \theta} |F(Q)|^2$$

$$I_{\text{el}} = I_0 \frac{\Omega}{4\pi} \frac{\sigma_{\text{XRF}}(E_0)}{\sigma_{\text{TOT}}(E_0)} \frac{1}{\sin \psi_1} \frac{C_{\text{el}} \alpha_{\text{el}}(E_0)}{\frac{\alpha(E_0)}{\sin \psi_1} + \frac{\alpha(E_1)}{\sin \psi_2}}$$



McXtrace



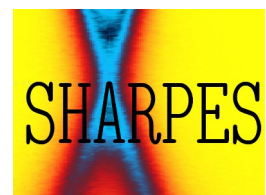
<https://mcxtrace.org/>

E. Knudsen *et al*, JAC **46**(3):679-696, 2013
A. Brunetti *et al.*, Spectrochimica Acta Part **B 59** (2004) 1725–1731
T. Schoonjans *et al.*, Spectrochimica Acta Part **B 66** (2011) 776–784

Before starting any new study from scratch, you should consider to inquire for existing structures and vibrational information from **databases** at:

- <https://alexandria.icams.rub.de/>
- <https://optimadeclient.materialscloud.io/> (catalog of catalogs)
- <https://www.materialscloud.org/home> (QE team)
- <https://nomad-lab.eu/nomad-lab/> (simulated structures)
- <https://next-gen.materialsproject.org/> (simulated structures)
- <https://www.crystallography.net/cod/> (sim and exp)
- <https://github.com/atztogo/phonondb> (phonopy db)





The SHARPES data base

- **Request from the ARPES community**
 - Collaboration with LPS/Orsay and Jussieu
- **Rationale**
 - Made to hold e.g. solid state spectroscopy data.
 - Search existing data.
 - Browse existing data.
 - Contribute Descriptive data (with login).
 - Future interface with other databases.



SHARPES Search | Service Hosting Angle-Resolved Photo Emission Spectroscopy

This page allows to search for entries in the SHARPES database. Specify any words to search for (see help below).

[Contribute](#) | [Browse](#)

SEARCH



[Help about search syntax](#)

[source code at gitlab](#) - (c) Synchrotron SOLEIL - E. Farhi, M. D'Angelo, A. Tejeda (2025).



Experiment UUID:
EXPERIMENT_2026-06-
01_ec8f1a06776bd2006a6feb33e730dee2b2c

Title: CuO slab

[Expand All Sections](#) | [Collapse All Sections](#)

References

- **UID:**
EXPERIMENT_2026-06-01_ec8f1a06776bd2006a6feb33e730dee2b2c205de01381e019f4b1c011a5da4
- **Data reference:** John Doe [farhi%40synchrotron-soleil.fr]
- **Publications:**
- **Publications:**

Owner

- Database: DB_sharpes
- Submitter: farhi
- Experiment PI: John Doe

Datasets

Experiment

- Type: laboratory measurement
- Date: 2026-06-01



Search results:

ec8f1a06	CuO slab / a copper oxide surface / John Doe	2026-06-01 submitted Mon Jun 1 15:25:51 2026 by farhi
EXPERIMENT	mesure%20de%20MgO / slab%2050%20nm%20sur%20du%20Si%20111 / farhi	submitted Wed Mar 25 10:16:17 2026 by farhi

Generated on Mon Jun 1 15:26:03 2026 by /usr/lib/cgi-bin/sharpes_search.pl

We inspire from **SSHADÉ** documentation (nomenclature, layout...) [<https://wiki.sshade.eu/>].

Source code: <https://gitlab.com/soleil-data-treatment/soleil-software-projects/sharpes-service>

Service : <https://data-analysis.synchrotron-soleil.fr/sharpes/>



The advent of A.I.
The end of equations and theory.

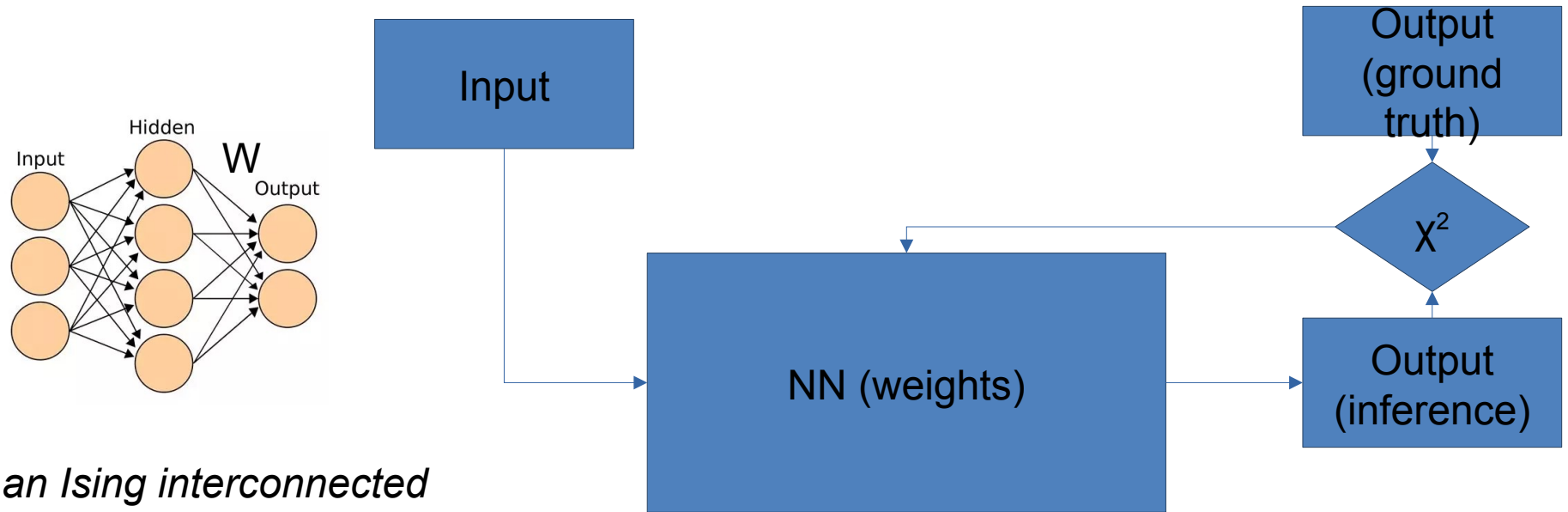
Reminder: Conventional neural networks

We have an **input**, an **output**, and some **internal 'weights'**. We optimize the NN internal weights (the 'memory') so that its output converges to the expected one.

This is a **pure optimization algorithm**.

The large number of parameters allows to approximate any function.

NN can NOT be exact as the training and usage implies random numbers.



Looks like an Ising interconnected spin-glass, huh ?

Reminder: Transformers / Self-Attention

Transformers rely on pure **linear algebra**.

We first 'embed' the **input** as *tokens* $\mathbf{x}$ (via quantization and PCA).
The **output** of the Transformer has the same size as the input $\mathbf{x}$.
The knowledge is stored into some Queries, Keys and Values **weights**.
Q.K.V vectors are determined from embeddings and **internal weights**

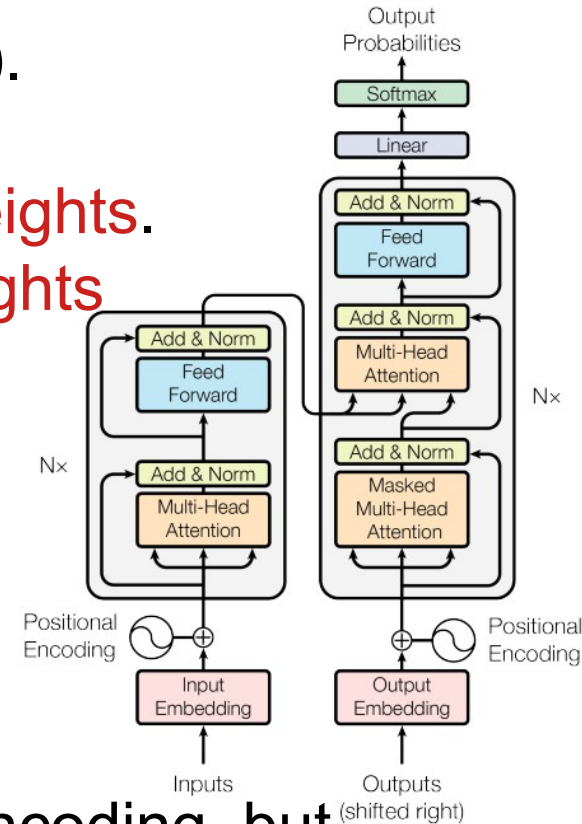
$$QKV = \mathbf{x} \cdot \mathbf{W}_{QKV}.$$

The weights are set during training.

The output is a dot product:

$$\mathbf{y} = \text{softmax}(\mathbf{Q} \cdot \mathbf{K}^T / \sqrt{d_K}) \cdot \mathbf{V} \quad [\text{softmax} = \text{logistic} = \text{Boltzmann}]$$

Transformers can NOT be exact as they rely on reduced latent encoding, but they are (nearly) **deterministic**.



See: <https://xmarva.github.io/blog/2025/building-a-transformer/>

Scientific method: Demonstrate



- 1) **Form a hypothesis** / Observe and Ask a Question
- 2) **Test** the Hypothesis (Experiment) e.g. Get sample (“real world”) and Measure
- 3) **Analyze Data** & Draw Conclusions
 - connect observables with the microscopic understanding.
 - “propagate errors” and understand the meaning of all parameters.
- 4) **Report** & Iterate

Methodology must be reproducible for cross-checking.

Data Processing strategies rely on tools that ***must*** derive from the scientific needs.

Where AI supports science, and where it deviates

✓ AI as a Scientific Accelerator (The Opportunity)

- **Data Processing:** Handles massive datasets far beyond human capacity.
- **Pattern Recognition:** Identifies complex correlations that may guide new hypotheses.

⬢ AI as a Methodological Divergence (The Caveat)

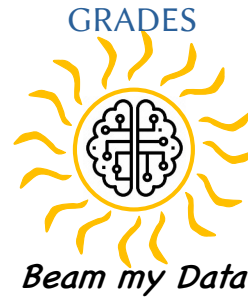
- **Opacity:** Many models are "black boxes"; the logical path to the answer is not explicit.
- **Falsifiability Gap:** Reliance on training data means AI outputs are restricted to that domain and are not inherently guaranteed to be demonstrable in a rigorous sense.
- **Justification:** AI predicts based on correlation (Input→Output), not necessarily on a proven causal mechanism required by the scientific method.



AI is best viewed as an advanced inference engine, requiring human scientists to apply the methodological rigor (testing, critique, causality) to validate its outputs.

By itself, AI does not follow the science path.

SELECTED A.I. APPLICATIONS AT SOLEIL



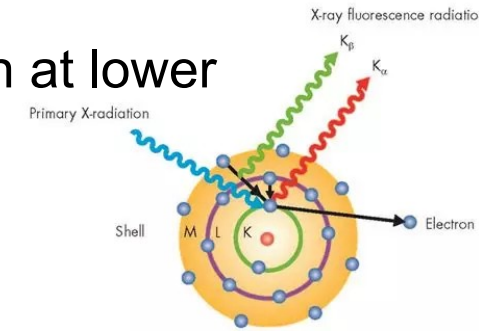
Hand-crafted methods can hardly generalise
one specific subject – one solution – short life-time.

We seek for methods which can **serve wider data types**, if possible.

Automatic data reduction

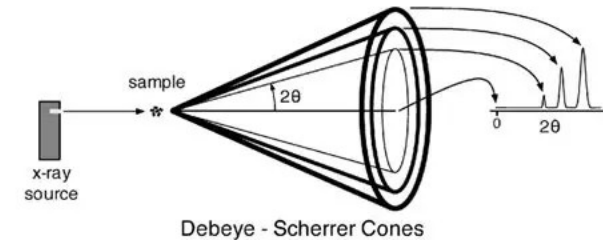
Most beam-lines are equipped with fluorescence detectors.

- Photons eject electrons in an absorption process.
- The holes are filled by an electron that “falls down” while emitting a 2^{dary} photon at lower energy.
- Generates a 1D energy spectra (e.g. 1-40 keV) [XRayLib]



Many beam-lines are equipped with diffraction 2D detectors.

- The crystalline periodic structure diffracts with Bragg law (coherent elastic scattering)
- Creates a 2D set of “rings” or “spots” vs angle [McXtrace/GSAS]



Other possible techniques:

- **absorption edges**: specific to atoms oxidation and coordination [FEFF/FDMNES]
- **small angle scattering**: specific to “large” structures (micelles, colloids, polymers, proteins, ...) [SASView]
- **denoising** (noise2void, noise2noise, DeepImagePrior, DeepDeWedge)

Automatic data reduction: XRF example

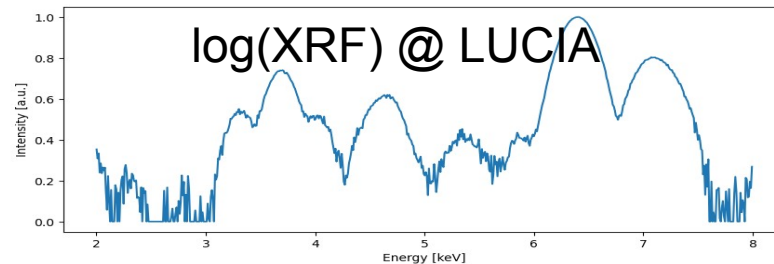
AI training requires **annotated data** (for supervised, quantified results)

No large annotated data sets. 😞

- **XRF data simulated** with McXtrace/XRayLib

KLM lines, Z=5-90, 40k samples.

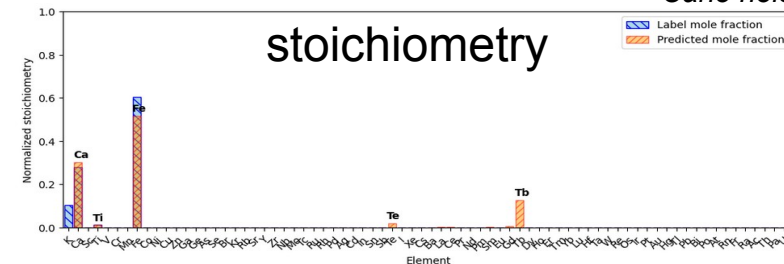
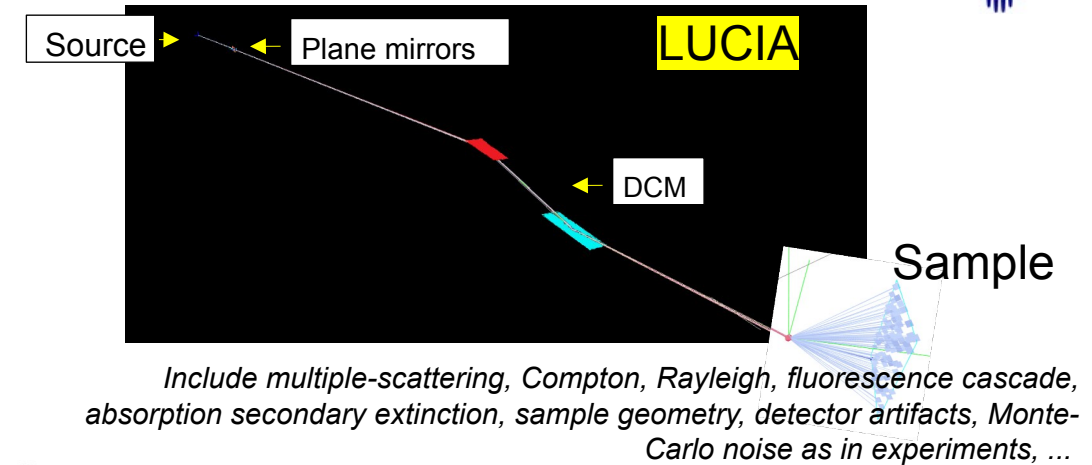
- We use a Visual Transformer + MLP.



Training (sim)

inference

- **Model can predict the stoichiometry of elements** on simulated spectra.
- The model is **flexible** on **energy range**, **resolution** and **number of points**
- **Rate:** 400k spectrum/s – **90% efficiency**.
- Future deployment on the BL → add metadata annotation/sample.



Work by
F. La Porta

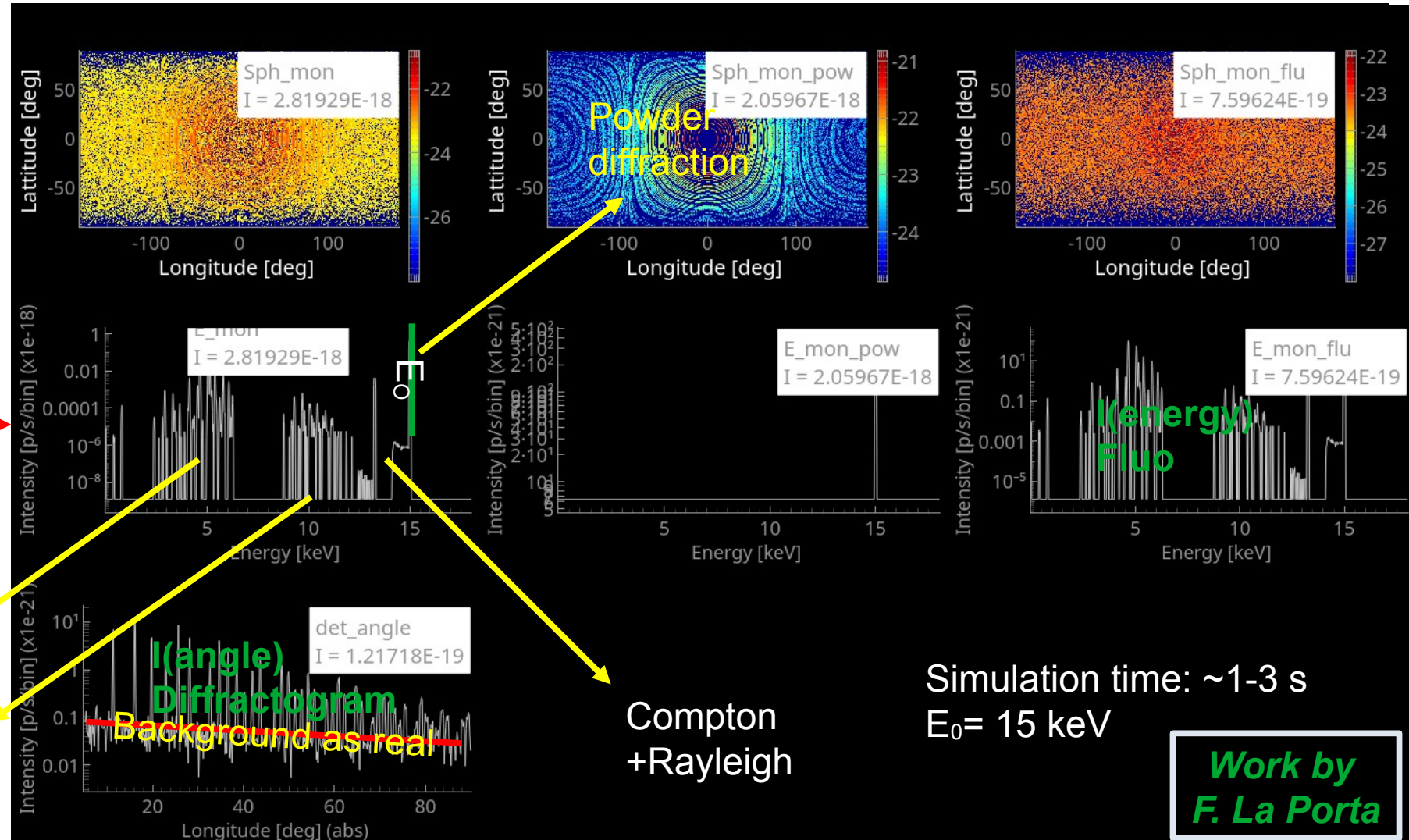
McXtrace

Automatic data reduction: XRD example

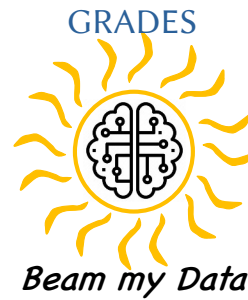
Next step: fast estimate of space-group from diffractogram.

Fluorescence is not really isotropic – appears as a background in angle. Can include e.g. air or sample container

CIF (structure)



A FUTURE : MATERIAL AGENTS



Enrich LLM's with many scientific tools

Agents everywhere

LLaMA

Today, LLMs are multimodal (MoE)

Built with Transformers and NN (**fully isolated**).

SOLEIL: We currently use a local Gemma4 / 3xRTX6000 Ada
1000 requests/h – 60% users via UI and 40% API.

Agents will soon be everywhere, no more coding

Agents execute actions, and re-inject results in loops, replace GUIs.

Our task:

- 1) Provide sovereign LLMs
- 2) Accept « secured » agents.



Conversations will replace user-interfaces.

Serves as GUI, e-logbook, and notebook. Lowers the gap for new users.

Each action is attached to e.g. a 'mini' context. When the LLM identifies it, the action is triggered and results are fed back to the conversation. Sequences of tool calls constitute 'Skills' (recipes).

« read the data and plot it », « fit the data with Lorentzians », « optimize the parameters », « compute a phase diagram for SrTiO₃ with QE ».

I have contributed to [llama.cpp](#)

Possible tools for synchrotrons:

- | | |
|--|--|
| <ul style="list-style-type: none">• Documentation search, including local knowledge bases.• Material modelling (ab-initio/classical MD).• Coding and testing.• Automatic data reduction.• Beam-line digital twin (virtual experiment). | <ul style="list-style-type: none">• Acquisition control (Tango control ⚠ / alignment).• Fill-in forms (SUNset/proposals).• Massive Data extraction and processing (from data bases). |
|--|--|

Giving ‘tools’ to LLM’s is just like a hammer to a baby.

WARNING to all opencode/claw/work users: you can’t guess what will happen. Whatever be the intelligence of an Agent, it requires sturdy tools.



Different solutions:

- Give a restricted list of tools.
- Give full access to a computer, and ask anything. This is VERY risky.
- Give full access inside a container/VM, and ask anything. Better but not certified.

What we are aiming for:

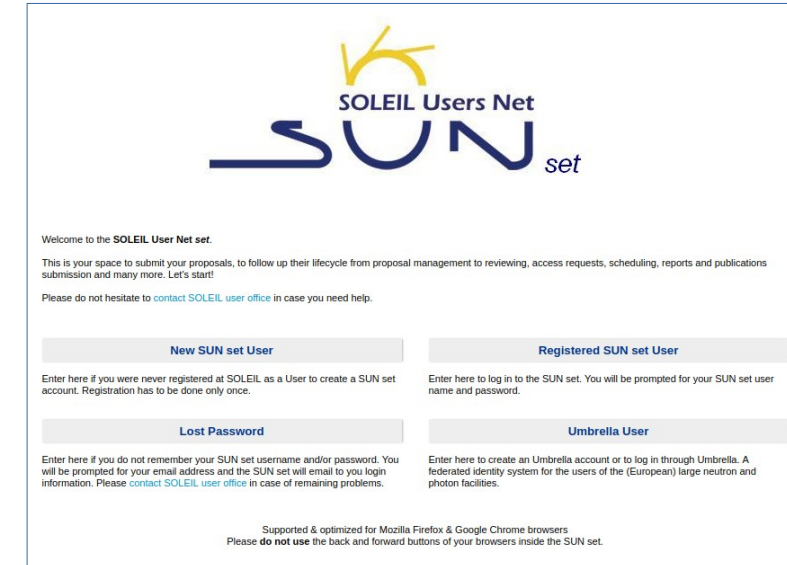
- A list of selected tools and recipes, inside a container.

Recipes will be assembled from code documentation and tutorials.

Tools need to be created and maintained.

How to write a proposal for beam-time at SOLEIL

- 1) Specify the **sample** you wish to study.
- 2) Specify the type of **measurement** you aim for:
 - Diffraction is first required. If known, indicate structure in proposal.
 - ARPES and RIXS are well suited for solid-state physics.
- 3) Identify relevant target **beam-line**, and subcommittee panel:
 - Structure and elemental analysis : DIFFABS, CRISTAL, ...
 - Dynamics : CASSIOPEE, ANTARES, GALAXIES, ...
- 4) Contact the BL **scientists** or <useroffice@synchrotron-soleil.fr>.
- 5) Write the scientific case, estimate measurement time (incl. Calibration and reference samples).
- 6) Justify the BL request (« *SOLEIL is the best ; that BL is really suited ; study never done before... »*)
- 7) Refine text in iterations with the BL scientist, well before the dead-line.
- 8) **Submit** at <<https://sun.synchrotron-soleil.fr/sunset/bridge/sunset/>>



Merci de votre attention

pepr-diadem.fr